MILITARY SPECIFICATION

CLOTH, COATED, NYLON, POLYURETHANE COATED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for three types of polyurethane coated cloth, heat, radio frequency or ultrasonically sealable with air holding characteristics (see 6.1).

1.2 Classification. The finished coated cloths shall be of the following types (see 6.2).

Type I - Plain weave, 4.5 ozs./sq.yd.
Type II - Plain weave, 5.5 ozs./sq.yd.
Type III - Oxford weave, 7.5 ozs./sq.yd.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging and Packing of Coated (Plastic, Rubber) and Laminated Fabrics.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFWAL/MLSE, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

FED-STD-595 - Color (Requirements For Individual Color Chips [3X5 Supplements])

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1487 - Glossary of Cloth Coating Imperfections.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents. The following Government document forms a part of this specification to the extent specified herein.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Application for copies should be addressed to the Federal Trade Commission, Washington, D.C. 20580.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article approval. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Materials.

3.2.1 Base cloth. The base cloth shall conform to the requirements of table I.

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TABLE I. Properties of the base cloth.

Characteristic	Requirement Type I	Type II	Type III
Yarn identification	nylon	nylon	nylon
Yarn size (denier)	210	210	200
Yarns per inch (min.)			
Warp	38	79	60
Filling	38	59	50
Cloth weight (ozs./sq.yd.)	2.25 ±0.1	4.0 ±0.1	3.0 ±0.1
Cloth weave	plain	plain	oxford

3.2.2 Coated cloth. The cloth shall be coated on one side only and shall conform to all the requirements of table II, when tested as specified in 4.4.5. At the supplier's option the coated side may be dusted with the dusting powder specified in 3.3.

TABLE II. Requirements of coated cloth.

Characteristic	Requirement Type I	Type II	Type III
Coating identification	polyether urethane	polyether urethane	polyether urethane
Coated cloth Weight (ozs./sq.yd.)	4.5 ±0.5	5.5 ±0.5	7.5 ±0.5
Breaking strength (min.)			
Warp (1 lb.)	170	320	200
Filling (1 lb.)	150	270	150
Tear strength (min.)			
Warp (1 lb.)	7	7	6
Filling (1 lb.)	7	7	3.5
Adhesion, coating to base cloth (min.) (lb./in.)	15	10	20
Air retention (allowable leakage)	None	None	None
Blocking (max.)	No. 2	No. 2	No. 2

N/A - Not Applicable

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3.3 Dusting powder. When used the dusting powder shall not interfere with the radio frequency or ultrasonic sealability of the fabric.

3.4 Color. The color of the coated fabric shall be specified by the procuring agency using MIL-STD-595, unless a standard color sample is provided (see 6.2).

3.5 Length and put-up. Unless otherwise specified, the coated cloth shall be put up on rolls as specified in PPP-P-1136 and shall be in one continuous piece of not less than 40 yards.

3.6 Width. The minimum width of the coated cloth shall be as specified (see 6.2).

3.7 Fiber identification. Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act as amended.

3.8 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of tests. The inspection and testing shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article submitted in accordance with 3.1 shall meet all the requirements of this specification for the type being certified (see 1.2).

4.3.1 First article inspection samples. The first article inspection samples shall consist of 1 linear yard, full width, of uncoated base cloth and 3 linear yards, full width, of the coated cloth upon which certification is desired (see 6.3).

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4.4 Quality conformance inspection.

4.4.1 Base cloth inspection.

4.4.1.1 Sample size. The sample unit shall be 1 yard, full width, of the uncoated base cloth, and shall be in accordance with table III. Each yard of material shall be from a different roll in the lot.

TABLE III. Sample size.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.1.2 Inspection. The base cloth shall be tested in accordance with table IV.

TABLE IV. Testing of base cloth.

Component	Characteristic	Requirement reference	FED-STD-191 test method
Base cloth	Material identification	3.2.1	1/
	Weave	3.2.1	Visual 2/
	Weight	3.2.1	5041
	Yarns per inch	3.2.1	5050

1/ A certificate of compliance shall be accepted for this characteristic.

2/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

4.4.2 Visual examination of end item.

4.4.2.1 Sample size. The sample size shall be in accordance with table V.

TABLE V. Sample size.

Lot size (yards)	Sample size (rolls)
Up to 1200 inclusive 1/	3
1201 to 3200 inclusive	5
3201 to 10,000 inclusive	8
10,001 to 35,000	13
35,001 to 150,000 inclusive	20
150,001 and over	32

1/ If the lot contains fewer than 3 rolls, each piece in the lot shall be examined.

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4.4.2.2 Inspection. The entire length of each roll called out in 4.4.2.1 shall be inspected on one side only. However, the side shall be alternated so that alternate rolls are examined on the face and back respectively.

The defects found shall be classified as listed below. In addition, the cloth shall be given a through-lighting inspection for pinholes. The through-lighting inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted, regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. The lot size shall be expressed in units of 1 linear yard. The sample unit for this examination shall be 1 linear yard. The Acceptable Quality Levels (AQLs) for the examination shall be 2.5 for major defects and 4.0 for total defects (major and minor) per 100 units (yards). The inspection level shall be level II.

	Classification	
Defect	Major	Minor
Uncoated side		
Any hole, cut or tear.	X	
Any slub, smash or multiple float.	X	
Baggy, ridgy or wavey cloth.	X	
Any strike through of coating.		X
Any spot or stain.		X
Coated side		
a. Any cut, hole, tear; any coated surface scratch or abrasion that results in baring the base cloth.	X	
b. Any surface scratch or abrasion that does not result in baring the base cloth.		X
c. Any area where coating is noticeably thinner.	X	
d. Any blister, tunnel or delamination of coating.	X	
e. Any lump or heavier coated area.		X
f. Foreign matter imbedded in coating.	X	
g. Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure.	X	
h. Coating cracked or checked.	X	

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Defect	Classification Major	Minor
Coated side (continued)		
i. Coating not applied over full width of base cloth.	X	
j. Any pinhole.	X	
k. Any light area or window resulting from improper distribution of pigment.	X	
l. Coating streak.		X
m. Any ripple, waviness or dimensional distortion.	X	
n. Any spot, stain or streak 1 inch or more in combined length and width dimensions.	X	
o. Any spot, stain or streak less than 1 inch in combined length and width dimensions.		X
p. Width less than minimum specified.		X
q. Color not as specified, off shade, uneven, mottled.		X
r. Any evidence of uncleanliness.		X
s. Any tackiness.	X	
t. Any odor other than that which is characteristic of the coating compound.		X

4.4.3 Examination for length and put-up.

4.4.3.1 Individual pieces. During the yard-by-yard examination of the cloth, each piece in the sample shall be examined for length and put-up. Any length found to be less than the minimum specified, or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. Any roll not in one continuous piece shall be considered a defect with respect to put-up. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length or put-up.

4.4.3.2 Total yardage in example. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

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4.4.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1136 to determine that packaging, packing and marking comply with section 5 requirements of this specification.

4.4.5 Testing of the end item. The methods of testing specified in table VI, shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for test purposes shall be 3 continuous yards full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be as indicated in table III.

TABLE VI. Test methods.

Characteristic	Requirement reference	FED-STD-191 test method
Coating compound polyurethane identification	3.2.2	1/
Coated cloth Weight	3.2.2	5041
Breaking strength	3.2.2	5100
Tearing strength	3.2.2	5134
Adhesion of coating	3.2.2	
Coating to coating		4.4.5.1
Air retention	3.2.2	4.4.5.2
Blocking	3.2.2	5872

1/ A certificate of compliance shall be accepted for this characteristic.

4.4.5.1 Adhesion - coating to coating. Ten specimens of the coated cloth shall be tested for adhesion. Each specimen shall be cut 1 inch (plus or minus 1/16 inch) wide and 6 inches (plus or minus 1/16 inch) long, with the length dimension in the warp direction of the coated nylon cloth. One of the specimens shall be superimposed and aligned on another of the specimens, with the coated sides in contact. The two specimens shall be heat sealed together, across the entire width, one-fourth inch from each end. The width of the seal shall be one-eighth inch (plus or minus 1/32 inch). A suitable inspection apparatus equipped with an autographic recording device (see test method

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5100 of FED-STD-191) shall be used to conduct the adhesion test. The front clamp of each jaw of the inspection apparatus shall be 1 inch by 3 inches. The rate of separation of the jaws of the inspection apparatus, under no load, shall be 12 inches (plus or minus one-half inch) per minute. One of the free ends of a specimen (an end that is not bonded) shall be placed in the lower jaw of the inspection apparatus. The other free end of the bonded specimen shall be placed in the upper jaw of the inspection apparatus. The bonded portion of the specimen shall be in the center between the two jaws. The jaws shall be separated until either breakage of the cloth or separation of the bond occurs. The highest value required to separate the bonded specimen or to break the cloth shall be recorded as the test result of the individual bonded specimens. This test shall be repeated until the other eight specimens have been tested for adhesion. The average value and lowest value obtained from the tests of the ten specimens shall be reported as the test results for the adhesion test. The average value shall be a minimum of 65 pounds per inch width, and the lowest value shall be no less than 45 pounds per inch width.

4.4.5.2 Air retention. Three 13-inch-diameter specimens of the coated nylon cloth shall be individually inspected for air leakage using an apparatus conforming to figure 1. The specimen with the coated side down shall be placed on the specimen holder. The rubber gasket shall be placed on the cloth side of the specimen. The plate collar shall be placed on the gasket. The specimen holder, the specimen, the gasket, and the collar shall then be securely bolted together so that a leak-tight fit shall result. A sufficient amount of water to keep the cloth-side surface of the specimen completely covered during the inspection shall be placed on the cloth side of the specimen. The specimen shall then be inflated with oil- and water-free air to a pressure of 10 ±½ pounds per square inch, gauge. This pressure shall be maintained for five minutes (plus or minus one-half minute). While this pressure is maintained, the specimen shall be observed for leakage through the top of the plate collar. Any air bubbles on the cloth side surface, which are produced by air pressure closing the spaces between the cloth and the coating, shall be removed. A continuous production (flow) of air bubbles is an indication of leakage and shall be cause for rejection. This test shall be repeated until the other two specimens have been tested for air leakage.

5. PACKAGING

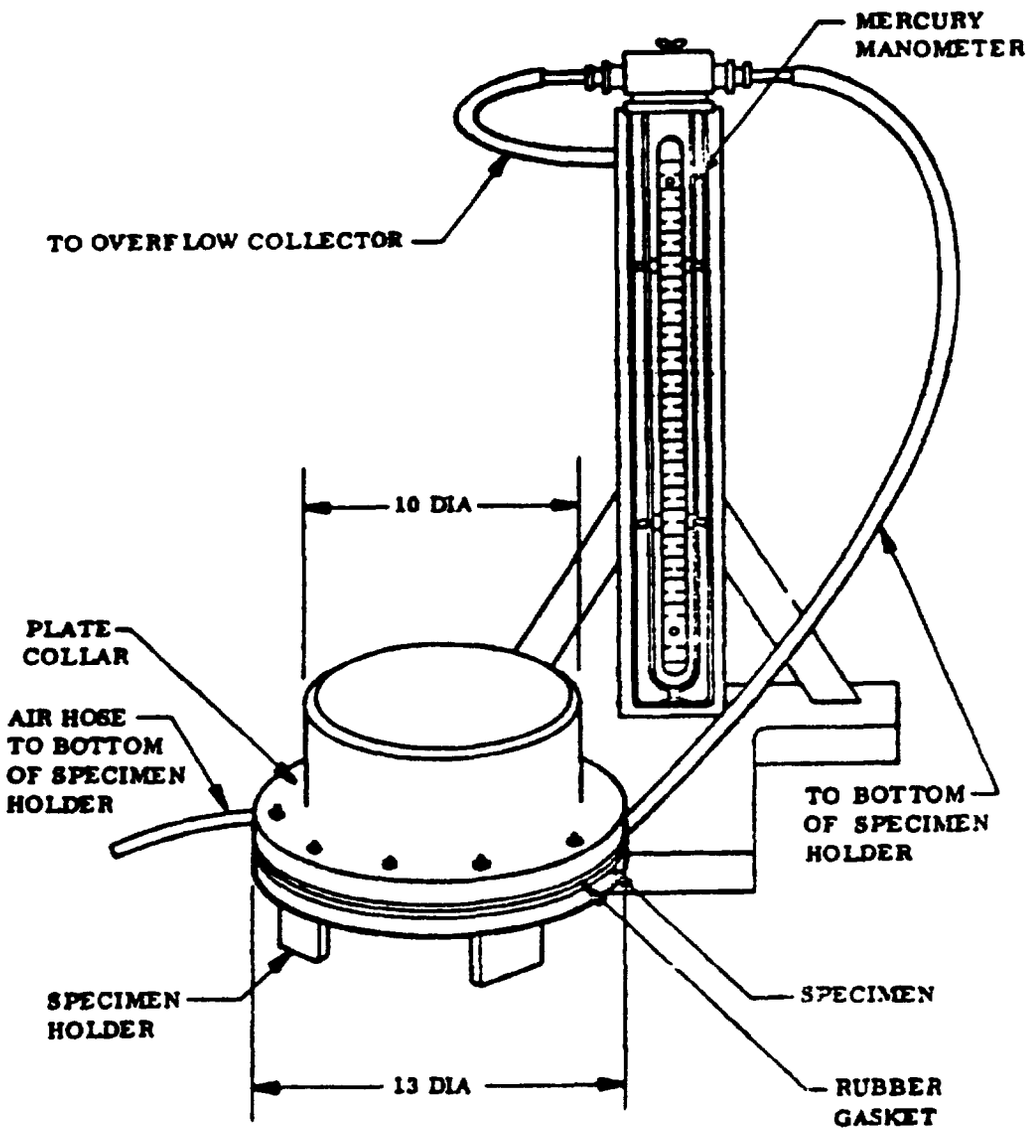
5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2 and 6.2.1).

5.1.1 Levels A and C. The cloth, put up as specified (see 3.5), shall be packaged in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

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DIMENSIONS IN INCHES

FIGURE 1. AIR LEAKAGE TEST JIG

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5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

6.1 Intended use. The coated cloth is intended for use in the manufacture of inflatable retarding devices for munitions.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. First article requirements (see 3.1).

c. Width required (see 3.6).

d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

e. Type required (see 1.2).

f. Color required (see 3.4).

6.2.1 Packaging options. In the preparation of contracts or orders, it must be noted that various options, choices and alternatives as indicated in PPP-P-1136 may be exercised in the preparation for delivery of the cloth.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample (or it may be a standard production item from the contractor's current inventory). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the first article in the document.

Custodian:
Air Force - 20

Reviewer:
Air Force - 18

Preparing Activity:
Air Force - 20

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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